Gym Management & Fitness Tracking System

Using Object Oriented Programming in C++

Presented By:

Muhammad Umair B(303
Isbah Rizwan (B-30322)
Saher Zaheer (B-30341)



Overview:

- Introduction
- Objectives and Key Features
- Modules and Functionalities
- Terminal View
- Conclusion

Introduction:

In today’s fast paced world, managing a gym manually becomes difficult and error prone.

This project solves this problem by providing a digital solution that handles trainer management, customer records, appointment scheduling, and workout tracking.

The system is developed using Object Oriented Programming concepts to make it structured, secure, and easy to maintain.



OBJECTIVES AND KEY FEATURES:

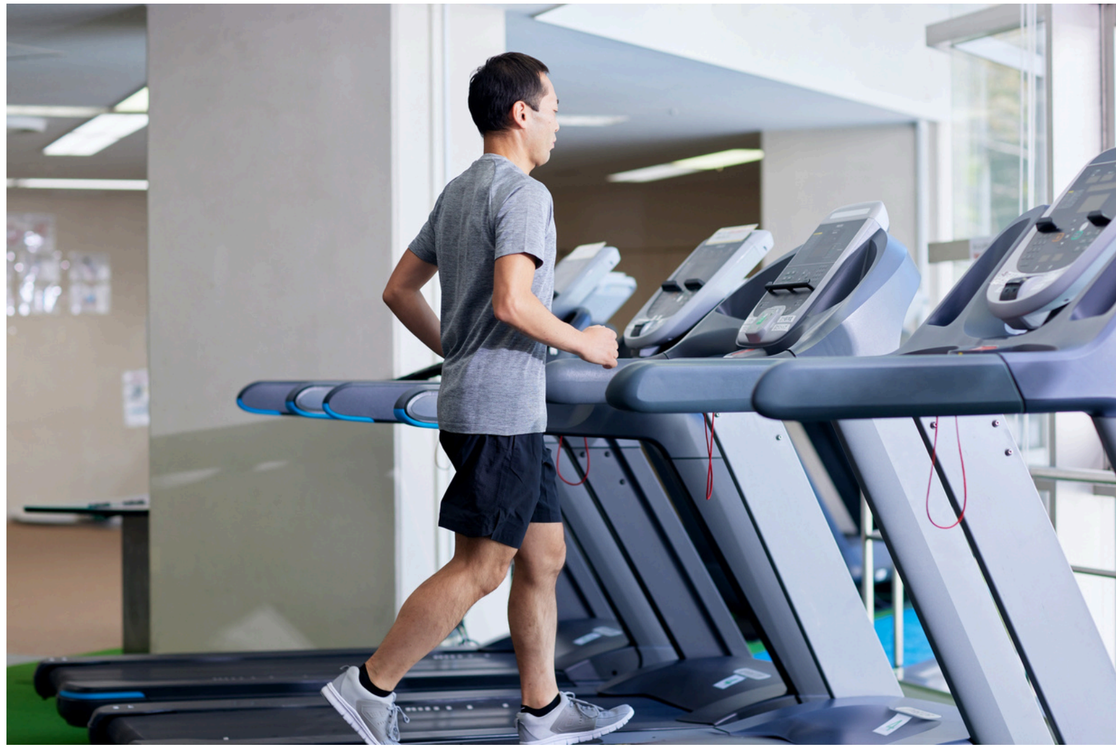
- Automate gym management tasks
- Provide secure login for all users
- Manage trainers and customers easily
- Enable easy appointment booking
- Allow trainers to assign workout plans
- Track customer workout history



Methodology :
(How it Works)

The Flow: "First, the Admin registers Trainers and Customers. Then, Customers book a time slot. Finally, Trainers assign workouts, and the system automatically calculates the health results."

Simple Logic: "We use 'In-Memory' storage, which means the program is very fast and doesn't need external files to run."



Key OOP Concepts:

Inheritance:
"We created one Parent workout class. All specific workouts like Cardio or Yoga inherit from it. This keeps our code clean and organized."

Encapsulation:
"We keep sensitive data like passwords and age private. They can only be changed through proper security checks."

Polymorphism:
"Even though every workout uses the same Calculate button, the math behind it changes. Boxing burns more calories than Yoga. The system is smart enough to know the difference."

Abstraction:
"We only show what the user needs to see. For example, the user clicks a button to calculate calories, but the complex logic behind it stays hidden. This makes the system easy to use and easy to manage."

Smart Features:

Time Check:

"The system checks every trainer’s clock. If a trainer is busy at 2 PM, the system will not allow another booking at that same time."

Date Safety:

"A trainer cannot give a workout plan for a date that hasn't happened yet. Everything follows a logical timeline."

Calculations:

Instead of a 'flat' calorie count, we use multipliers for 'High' or 'Low' intensity to give members a realistic view of their progress."



Conclusion:

"This project shows how we can use smart programming to solve real gym problems. It is fast, secure, and very easy for both trainers and customers to use."



Thank You

Any Questions?